

Erdős Problem #340

EP-340

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $A = \{1, 2, 4, 8, 13, 21, 31, 45, 66, 81, 97, \dots\}$ be the greedy Sidon sequence: we begin with 1 and iteratively include the next smallest integer that preserves the Sidon property (i.e. there are no non-trivial solutions to $a + b = c + d$). What is the order of growth of A ? Is it true that $|A \cap \{1, \dots, N\}| \gg N^{1/2-\epsilon}$ for all $\epsilon > 0$ and large N ?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>